

Guidelines for the In-House Production of Reference Materials

Version 2

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*Setting standards
in analytical science*

Guidelines for the In-House Production of Reference Materials

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Executive Summary

These guidelines were originally produced under DTI's 1994-1997 VAM programme to address the need for a set of easily understandable guidelines for the in-house production of reference materials.

The guidelines highlight the key issues that analysts should take into account when preparing reference materials in-house. These include material selection and preparation, homogeneity and stability testing and measurement of the reference value(s). More detailed information and guidance on the important points is given, and these are illustrated with a range of worked examples. The user is also provided with references to additional relevant information such as ISO Guides.

The use of reference materials is becoming increasingly important in analytical quality control to help laboratories monitor their day to day performance. These guidelines provide practical advice and information to help analysts prepare their own materials more effectively.

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1. SCOPE

The aim of this document is to provide guidance to analysts on how to prepare in-house reference materials. Although the term 'Reference Material' is defined in ISO Guide 30 (see Section 3), these guidelines are just as appropriate for materials often referred to as:

- working reference material
- check material
- quality control standard
- secondary working standard

In other words these guidelines are applicable to those materials that are produced in-house in order to:

- ❖ monitor instrument and method drift
- ❖ calibrate or assess the performance of an analytical system

This document is not intended to provide fully prescriptive methodology for producing the materials, but to:

- ❖ raise the issues that should be considered
- ❖ signpost the analyst to the appropriate official standards and/or guides
- ❖ provide practical examples where appropriate

It is beyond the scope of this document to provide guidance on analytical techniques and thus it has been assumed that a reasonable degree of analytical competence exists in the establishment preparing the reference material.

2. BIBLIOGRAPHY

This document is intended to act as a signpost to the appropriate standards, guides and other references. At the end of each section all relevant references will be listed in full. However a full list of references is given below (details on how to obtain these references are given in Appendix A):

ISO Guide 30, (1992) : Terms and definitions used in connection with reference materials

ISO Guide 31, (1981) : Contents of certificates of reference materials

ISO Guide 33, (1989) : Uses of certified reference materials

ISO Guide 34, (1996) : Quality system guidelines for the production of reference materials

ISO Guide 35, (1989) : Certification of reference materials - General and statistical principles

ISO 5725-2, (1994) : Accuracy (trueness & precision) of measurement methods - Basic method for determination of repeatability and reproducibility for a standard measurement method

BS 5309 Part 1, (1993) : Methods for Sampling Chemical Products - Introduction and general principles

BS 5309 Part 2, (1993) : Methods for Sampling Chemical Products - Sampling of Gases

BS 5309 Part 3, (1993) : Methods for Sampling Chemical Products - Sampling of Liquids

BS 5309 Part 4, (1993) : Methods for Sampling Chemical Products - Sampling of Solids

European Commission Document, BCR/48/93 (Dec 1994) : Guidelines for the production and certification of BCR reference materials

National Institute of Standards and Technology Publication 260-100 (1993) : Standard Reference Materials - handbook for SRM Users

International Harmonised Protocol for Proficiency Testing of (Chemical) Analytical Laboratories - Thompson & Wood : Journal of AOAC International : Vol.76, No.4, 1993

General Principles of Good Sampling Practice - Crosby.N.T & Patel.I

VAMSTAT (1994) - a computer based training package on statistics for analytical chemists

Note : This list is not exhaustive and only the more generic type documents are listed i.e. those that are applicable to a wide range of analytical sectors. Within specific analytical sectors other official standards/guides exist which would be of value e.g. within the petroleum industry one particular reference would be :

ISO 4259, (1992) : Petroleum products - Determination and application of precision data in relation to methods of test.

3. DEFINITIONS

All the definitions listed in this section have been taken from ISO Guide 30, but to assist analysts further additional comments/notes are given in italics.

Reference Material (RM) : Material or substance one or more of whose property values are sufficiently homogenous and well established to be used for the calibration of an apparatus, the assessment of a measurement method or for assigning values to materials.

Note : for the purposes of this guidance document this will include such terms as working reference materials, check materials, quality control standards and secondary working standards. It should also be noted that many RMs are accompanied with a certificate or other documentation, but this should not be taken to mean that they are a true CRM, or even equivalent to a CRM in the context of the ISO definition of a CRM below.

Certified Reference Material (CRM) : Reference material, accompanied by a certificate, one or more of whose property values are certified by a procedure which establishes its traceability to an accurate realisation of the unit in which the property values are expressed, and for which each certified value is accompanied by an uncertainty at a stated level of confidence.

Homogeneity : Condition of being of uniform structure or composition with respect to one or more specified properties. A reference material is said to homogeneous with respect to a specified property if the property value, as determined by tests on samples of specified size, is found to lie within the specified uncertainty limits, the samples being taken either from different supply units (bottles, packages, etc.) or from a single supply unit.

Stability : Ability of a reference material, when stored under specified conditions, to maintain a stated property value within specified limits for a specified period of time.

Sample : Representative quantity of material extracted (*taken / selected*) from a batch of reference material.

Accuracy : The closeness of agreement between a test result and the accepted reference value.

Precision : The closeness of agreement between independent test results obtained under prescribed conditions.

Repeatability (of results of measurements) [*Precision under repeatability conditions*] :
Closeness of the agreement between results of successive measurements of the same measurand carried out subject to the following conditions:

- the same measurement procedure
- the same observer
- the same measuring instrument, used under the same conditions
- the same location
- short time interval

Reproducibility (of results of measurements) [*Precision under reproducibility conditions*] : Closeness of the agreement between the results of measurements of the same measurand carried out subject to the following conditions:

- the same measurement procedure
- different measuring instruments
- different locations
- different conditions of use
- over a relatively long time period



ISO Guide 30, (1992) : *Terms and definitions used in connection with reference materials*

4. INTRODUCTION

Reference materials (RMs) which are produced in-house (often referred to as working reference materials, check materials, quality control standards or secondary working standards) are required for use by laboratories on a day to day basis. The attraction of using such RMs is that they provide a less expensive option than using certified reference materials (CRMs) and can closely resemble the laboratory's routine test samples. However, it must be stressed that they **do not** replace the use of CRMs, but merely enable CRMs to be used on a less frequent basis. It is of paramount importance that the RMs are appropriately prepared to ensure that they are fit for the purpose for which they are intended. Therefore, this document is aimed at giving guidance to those who wish to prepare RMs in-house. It does not however provide a fully prescriptive set of instructions but instead aims to raise the various issues that need to be considered.

Before producing RMs in-house the following points should be taken into account:

- ❖ RMs can be used in day to day quality control systems.
- ❖ RMs should not be used as a replacement for CRMs for **fundamental purposes** (where traceability and a full uncertainty budget are required) such as :
 - method development and validation
 - verification of the correct application of standardised methods
 - calibration and verification of the measurement process
- ❖ RMs should wherever necessary be calibrated against a CRM or certified measuring device or instrument being used, when the RM is first prepared and thereafter on a regular basis

Note: When a new batch of the RM is prepared as well as referencing it as mentioned above it should also be compared to the batch of the material being replaced.
- ❖ The decision on whether a RM is fit for its intended purpose and the frequency of checking its assessed value(s) needs to be carefully considered by the analyst depending on :
 - intended use of the RM
 - level of homogeneity
 - stability of the material
 - precision of the method for which the RM will be controlling
 - frequency of analysis

Each subsequent section of this document considers the various stages in the preparation of RMs in-house and their use. A summary of these stages is shown in Figure 1, below. Depending on the particular situation it may not be necessary to undertake each stage of the process, for example :

- the material selected may already be in a suitable form and thus the 'material preparation' stage may therefore be omitted.
- historic or published data may exist on the homogeneity of the material selected which may negate the need for the 'homogeneity testing' stage.
- by the nature of the material coupled with data already available the stability of the material may already be established and therefore the 'stability testing' stage may not be required.
- if the RM is to be used for checking the day to day control of a method and its homogeneity and stability is well established it may not be necessary to obtain an accurate reference value.
- by the nature of the way the material is prepared it may not be necessary to actually measure the reference value e.g. for a gravimetrically prepared aqueous metal standard, the metal concentration can be calculated.

Figure 1 illustrates the various paths that can be taken in the production process. **However, it is strongly recommended that before any particular stage in the process is omitted the appropriate section in this guide is carefully read to be sure that its exclusion is appropriate.** For example:

- even though you may have selected a source of well prepared material and therefore intend to omit the 'material preparation' stage, appropriate labelling of the RM is important.
- homogeneity can change from batch to batch of readily available material e.g. a food material and therefore before omitting the 'homogeneity testing' stage you need to be sure that any likely changes in homogeneity will not be significant enough to make your RM invalid. In addition, whilst the bulk material may appear homogeneous, it is necessary to ensure that the individual units produced from the bulk are also homogeneous
- stability data may be available on a given material, but this data may be based on the pure substance and if the RM being prepared has even small levels of impurities its stability can be significantly different.



ISO Guide 30, (1992) : *Terms and definitions used in connection with reference materials*



ISO Guide 33, (1989) : *Uses of certified reference materials*



National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*

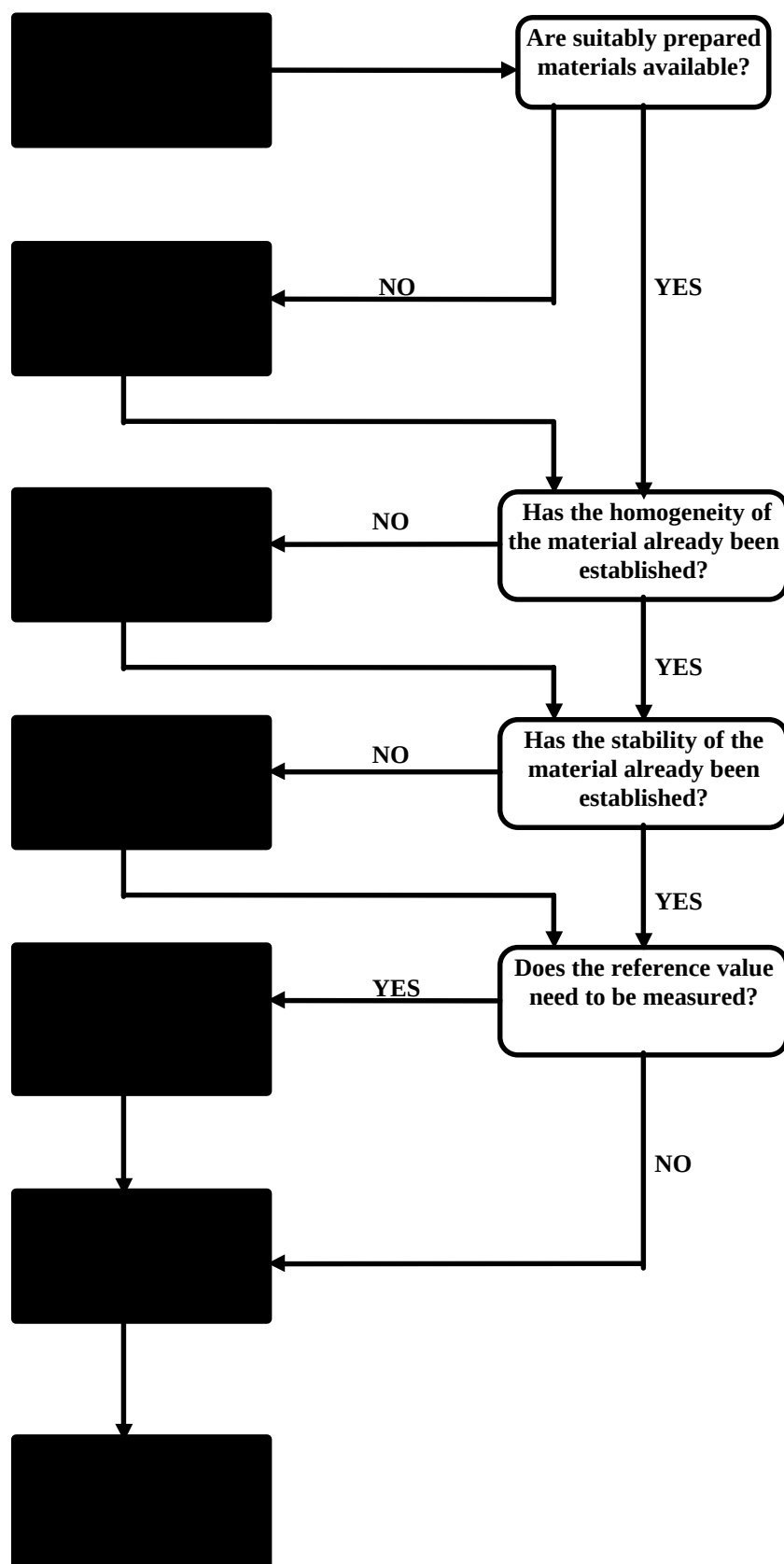


Figure 1: FLOW CHART OF STAGES FOR THE IN-HOUSE PRODUCTION OF RM's
5. MATERIAL SELECTION

The ease of selecting a suitable material will depend on the nature and type of RM being produced. For single substance RMs such as a pharmaceutical compound, pesticide, etc. there is likely to be a readily available source of material. For liquid matrix RMs such as an aqueous metal standard, a pesticide solution etc. it may well be possible to produce the RM by preparing it gravimetrically from readily available materials. For solid matrix RMs or liquid matrix RMs containing suspended matter, **the RM needs to represent the matrix of the samples that are being analysed by the method for which the RM is intended to control.** In these cases the identification of a suitable source may be very difficult for some matrix materials e.g. sewage sludge or less so for those commercially available e.g. white bread flour. In some cases it may be possible to use excess material from a proficiency testing scheme

When identifying a suitable source of material the following issues should be considered:

- ❖ Matrix of interest
- ❖ Properties/analytes of interest
- ❖ Appropriate concentration levels or physical values
- ❖ Intended end use
- ❖ Degree of homogeneity required
- ❖ Degree of stability required (shelf life)
- ❖ Quantity of material required
- ❖ Suitability of materials already available
- ❖ Ease of sourcing and sampling
- ❖ Available storage facilities
- ❖ Health and safety implications

Clearly for the more complex matrix materials where data on their composition, homogeneity, stability etc. is often not available it will be necessary to carry out some feasibility studies to identify suitable sources of material. For some materials, for example industrial products (e.g. pharmaceuticals, pesticides etc.), data will often be readily available.



ISO Guide 35, (1989) : *Certification of reference materials - General and statistical principles*



European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials*



National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*

6. MATERIAL PREPARATION

The material should be prepared/processed in such a way as to ensure that its final form is fit for its intended use and that it is sufficiently stable. The amount of material to prepare depends on the ideal sample size, stability of the material, storage capacity, the frequency of its use and the processing equipment available.

The procedures for processing the material need to be carefully chosen in order to achieve the required homogeneity and maximise the materials shelf-life. Typical operations include drying, grinding, sieving, filtering, mixing, rifling, purification, sterilisation etc. Care should be taken in selecting a suitable processing procedure in order to prevent any change or degradation in the form and/or concentration of the analytes of interest.

Example :

If the reference material is to be in the form of a dried material, consideration needs to be given to the advantages and disadvantages of oven drying, freeze drying, air drying, vacuum drying etc., e.g. for temperature sensitive materials and those containing volatiles.

Example :

Sterilisation by gamma irradiation does provide a good method of maximising the shelf-life of a material. However, whilst this technique would be ideal for RMs that are intended for elemental analysis, it is not suitable for all analytes. For example, some organo-tin compounds, pesticides and pharmaceuticals may decay on being subjected to gamma irradiation. Indeed, some pesticides are designed to break down when subject to irradiation.

Example :

For a single substance RM such as an organic compound, the material is likely to be readily available but it is important to check its purity and if necessary undertake purification of the material. The purity data may already be available from the manufacturer of the material, but care must be taken when interpreting this data. For example the purity may have been calculated by subtracting the impurities from 100% to obtain the quoted purity, but this is only accurate if all the impurities have been identified - see Section 9 for further details. In carrying out this type of procedure, alternative analytical approaches need to be considered in order to ensure the reliability of the assigned purity value.

Example :

Milling and mixing can frequently introduce impurities, particularly when using mechanical equipment, unless the equipment has been rigorously cleaned and checked in accordance with well-defined quality assurance procedures.

At this stage careful consideration needs to be given as to whether the material should be kept in bulk form, or subdivided into a more appropriate number of units. The disadvantages of one bulk sample is that repeated sub-sampling from one container carries the risk of degradation or contamination. If degradation or contamination of the material is likely the material should be subdivided into smaller batches and stored in appropriate containers in order to maintain the integrity of the material. The choice of the size of each unit is dependent on the intended end use of the RM. An assessment of homogeneity and stability should then be conducted on a representative number of these individual units. However, if a large batch of material has been prepared it is recommended that this is initially checked for homogeneity before subdividing into smaller units and carrying out the full homogeneity assessment.

Example :

A material that is prone to oxidation in the atmosphere or is volatile should be packaged as single shot units. It should also be borne in mind that many solvents are volatile and the concentration of a material dissolved in such a solvent may well change unless appropriate precautions are taken.

Example :

A soil RM may be produced by processing 10 kg of raw material and subdividing into 200 units each of 50g. However, the processing of the soil can often be quite difficult in terms of achieving a homogeneous material and it may need to be re-processed i.e. further grinding, mixing sieving etc. many times. Therefore, the bulk material should be checked for homogeneity after processing so that it can easily be re-worked if the level of homogeneity required has not been achieved. **Only when the homogeneity check on the bulk proves to be adequate should the material be subdivided and the full homogeneity assessment carried out on the individual units.**

The choice of container, its closure, the atmosphere inside it and the storage conditions necessary to maintain the integrity of the RM should be carefully considered:

- ❖ bottles should be sealed with a suitable closure
- ❖ vials should be sealed or equipped with stoppers
- ❖ sachets should be double sealed

Example :

A metal spectrometric solution should be placed in a **suitable plastic**, rather than glass, container since metal ions are more prone to adsorption onto glass. Nevertheless, leaching experiments should be carried out on the plastic container using the blank acid solution to determine its long-term suitability.

Example :

Liquid matrices that need to be stored at temperatures of -20°C or lower are not suitable for glass containers if they will freeze during storage, since the glass containers may be prone to cracking.

Example :

Materials known to be sensitive to air should be packed in an inert gas atmosphere, into sealed glass or hermetically welded metal containers.

Appropriate labelling is essential. Wherever possible each container should include the following, as well as complying with any other quality system requirements:

- ❖ a unique reference name or number
- ❖ a batch and sample number
(this may be incorporated into the reference number)
- ❖ reference or nominal value
- ❖ weight or volume of contents
- ❖ any safety warning signs
- ❖ storage requirements e.g. temperature, humidity, light etc.
- ❖ date of expiry or re-test

Example :

Reference Material

RM999 - Fish tissue
Mercury Concentration : ~ 5 mg/kg
Batch No: 001 Sample No : 001
Contents : 5g
Store at 4°C
Date of Expiry : 25 Dec 1997

Whilst for clarity it would be better if the label was printed or typed, if such facilities are not available than a clearly hand written label should be used. In terms of safety warning signs small pre-printed labels are commercially available from a number of outlets.

-  **ISO Guide 34, (1996) : *Quality system guidelines for the production of reference materials***
-  **European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials***
-  **National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users***

7. HOMOGENEITY TESTING

Depending on the RM being produced there may well be data already available that provides you with the assurance that the material being used is homogeneous - however you should read the whole of this Section to ensure that the data available does indeed fully meet the requirements.

The RM should be homogeneous, **that is the difference between representative sample measurements should be smaller than the overall uncertainty limits of the measurements**. Thus it is necessary to assess whether any variation between representative samples is fully accountable to the uncertainty of the measurement method being used (a full definition is given in section 3). The object of the test for homogeneity is mainly to determine the distribution at a constant level of any impurities e.g. moisture, interference's or irregularities e.g. incomplete dissolution. The degree of homogeneity is dependent on the quantity of material taken i.e. **a RM can only be said to be homogeneous at or above the weight/volume of the representative samples analysed**. Thus, the quantity of material chosen for testing should not be greater than that used in the routine analysis for which the RM will be employed.

Example :

In preparing a material for use in calibrating a scanning differential calorimeter, it will be necessary to ensure that the material is homogeneous for samples as small as 2-3mg. Alternatively, if a material is being produced for an analysis which requires 5g of sample, then homogeneity at the 3-4g level will suffice.

As mentioned in Section 5, the homogeneity testing should be carried out on representative samples from the final packed units, but it is prudent where a bulk material (i.e. the total quantity of material prior to its sub-division - which may vary from a few grams, for example, to many kgs or even tonnes) has been obtained to carry out preliminary homogeneity testing prior to subdividing into smaller units i.e. homogeneity checks form an integral part of the material preparation stage.

When testing for homogeneity the following issues need to be considered in the context of the degree of homogeneity required: the material to be tested, the parameters being measured, the quantity of the RM required to undertake an analysis and the intended use of the RM.

(a) Selection of homogeneity test(s): the most important consideration in the selection of the test(s) to be used is its precision, since at this stage accuracy is less important. Thus, whilst the analytical method(s) used for the measurement of the reference value (see section 9) will be selected for its accuracy and traceability, there may well be other methods that can be used for the homogeneity testing which are more convenient to use and have good repeatability. **It is important that all relevant methods be used to detect for all possible inhomogeneity.**

Example:

In the preparation of an RM for the determination of sulphur by neutron activation analysis, it would be appropriate to use X-ray Fluorescence Spectroscopy for the homogeneity testing since this technique has the appropriate degree of precision.

Example:

In the preparation of a single organic compound RM the homogeneity may well be assessed by the determination of the purity using high performance liquid chromatography (HPLC), gas-liquid chromatography (GLC) or differential scanning calorimetry (DSC). However, such techniques will not determine the moisture content and thus a technique such as Karl-Fischer would also need to be used for the homogeneity assessment - see Section 9 for more details. Problems due to the possible presence of volatile solvents also needs to be considered .

Where feasible, the measurements on all samples should be carried out by the same operator, against the same calibrants, on the same instrument(s), within the same analysis run to ensure the optimum conditions for repeatable measurements.

(b) Sample selection: this is normally carried out by either random or systematic selection, normally on the final packed units (this may be one bulk or many individual units).

Example:

A soil RM would be expected to have potential inhomogeneity down to very small sample sizes due to the nature of the material. It is therefore prudent to carry out a **random** selection of individual units and sub-samples from each unit in order to assess both within unit and between unit homogeneity. Clearly, if the individual units are single shot samples (i.e. all the sample is used for the analysis), the within unit homogeneity is not important.

Example:

A glass RM may have been produced by cutting small units from a large block of manufactured glass. In this case it is clear that any inhomogeneity may well be caused by the position of the unit within the glass block i.e. in the middle, near the edge, etc. In this case it would be prudent to **systematically** select samples from specific positions since random selection could for example exclude all samples taken from the edge of the block. If multiple blocks have been used, the homogeneity for between block also needs to be assessed, this time using **random** selected samples.

Example:

A RM which is a clear solution would not normally need to be assessed for within unit homogeneity since inhomogeneity is eliminated by simple manipulations such as shaking the container before use. However, the possibility of particulate matter which is not visible should be taken into consideration.

Example :

Homogeneity data may already exist for an organic material which is being used to produce a RM. However, if the bulk material is to be subdivided into a number of individual units, although homogeneity data may exist on the bulk material if there is potential during the subdividing for differential contamination. In this case you would **systematically** sample a unit from every 'x' units packaged in order to assess any increasing trends in the level of contamination.

(c) Number of samples to select: there are no definitive rules that can be given since it does to a large degree depend on the type of material, the preparation process, the end use and your organisation's own QA requirements. However, the following is a good guide:

- for assessing between unit homogeneity it is recommended that the following formula (taken from the British Standard on methods for sampling chemical products - BS5309 Part 1) is used:

the next highest integer to three times the cube root of the number of individual units

It is recommended that not less than ten units should be used.

- for assessing with-in unit homogeneity it is recommended that about 5 or 6 sub-samples are each taken from 2 or 3 units.
- for a single bulk material it would be prudent to calculate the number of potential analysis units and then apply the BS5309 rule.

Example:

For a soil sample that has been prepared as a bulk of fine powdered material and subdivided into one thousand 50g units, the following would be carried out:

- the between unit homogeneity would be tested by analysing
 $3 \times \sqrt[3]{1000} = 30$ units
- the with-in unit homogeneity would be tested by analysing say five 1g portions from each of three individual units.

Example:

For a pesticide RM that is to be kept as a bulk consisting of 100g it is first necessary to assess the smallest quantity of the material that will be used for any one analysis.

Assuming that 0.1g will be used, the following would be carried out:

- the number of analysis units would be calculated
i.e. $100 / 0.1 = 1000$
- $3 \times \sqrt[3]{1000} = 30$ sub-samples would be taken for the homogeneity testing.

However, if the RM is to be used in 0.25g units then:

- number of analysis units = $100 / 0.25 = 400$
- $3 \times \sqrt[3]{400} = 23$ sub-samples would be taken for the homogeneity testing.

Example:

For a bulk water RM (which may contain fine particulate matter) of 10 litres, from which 50ml sub-samples will be used for each quality control check, the following would be carried out:

- the number of analysis units would be calculated
i.e. $10000 / 50 = 200$
- $3 \times \sqrt[3]{200} = 18$ sub-samples would be taken for the homogeneity testing.

Example :

For a metal RM which consists of 5cm² plates cut from ten metal sheets each 100cm x 100cm, the following would be carried out:

➤ the with-in sheet homogeneity would be assessed by dividing one of the metal sheets into a 25cm by 25cm grid (i.e. 16 segments) and selecting randomly one metal plate cut from each of these segments. This would be repeated for 2 or 3 sheets.

➤ the between sheet homogeneity would be assessed by applying the BS5750 formula to the total number metal plates

i.e. ➔ total number of plates - $[10 \times (100 \times 100) / (5 \times 5)] = 400$
 ➔ $3 \times \sqrt[3]{4000} = 48$ sub-samples would be taken (randomly) for the homogeneity testing.

(d) Sample Size: the quantity of material used to assess the homogeneity should not be greater than the quantity that will be used when subsequently using the RM for calibration or validation purposes.

(e) Statistical tests to apply: it is beyond the scope of this document to give full mathematical formulae for the individual statistical tests to be applied. However, such formulae are readily available in numerous texts on statistics including a number of the references given below. The basic concept is that the statistical tests need to assess whether there is any statistically significant variation between the range of measurements made on the RM and the uncertainty of the measurement method being used. All measurement results should be included since the elimination of statistical outliers would defeat the object of the exercise.

A recommended test is the F-test which should be used to :

- compare the standard deviation of the measurement results of the RM with the method repeatability, obtained by replicate analysis (the number of replicates depending on the precision of the method - if this is not known at least ten should be used) on one of the samples (this is only applicable if a non-destructive measurement technique is used).

- compare the relative standard deviation of the measurement results of the RM with the method repeatability, obtained by replicate analysis (the number depending on the precision of the method - if this is not known at least ten should be used) on a CRM (this is applicable if a destructive measurement technique is used). **Care should be taken since if the CRM is very dissimilar to the RM the repeatability may not be representative.**

- compare the relative standard deviation of the measurement results of the RM with a quoted method error (repeatability) published for the measurement method used.

The degree of homogeneity required for a particular RM will depend on its intended purpose, but it is normal to conduct the F-test at the 95% or 99% confidence level.



ISO Guide 31, (1981) : *Contents of certificates of reference materials*



ISO Guide 34, (1996) : *Quality system guidelines for the production of reference materials*



ISO Guide 35, (1989) : *Certification of reference materials - General and statistical principles*



ISO 5725-2, (1994) : *Accuracy (trueness & precision) of measurement methods - Basic method for the determination of repeatability and reproducibility for a standard measurement method*



BS 5309 Part 1, (1993) : *Methods for Sampling Chemical Products - Introduction and general principles*



BS 5309 Part 2, (1993) : *Methods for Sampling Chemical Products - Sampling of Gases*



BS 5309 Part 3, (1993) : *Methods for Sampling Chemical Products - Sampling of Liquids*



BS 5309 Part 4, (1993) : *Methods for Sampling Chemical Products - Sampling of Solids*

-  **European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials***
-  **National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users***
-  **International Harmonised Protocol for Proficiency Testing of (Chemical) Analytical Laboratories**
-  **General Principles of Good Sampling Practice**
-  **VAMSTAT (1994) - a computer based training package on statistics for analytical chemists**

8. STABILITY TESTING

Depending on the RM being produced there may well be data already available that provides you with the assurance that the material being used is stable for the intended time span - however you should read the whole of this Section to ensure that the data available does indeed fully meet the requirements.

The aim of stability testing is to determine whether the RM maintains all its reference values from the time of production until the moment it is used. **The frequency of stability testing depends on the risk of any change in the reference value with time.** When selecting the material from which to produce the RM it is beneficial if stability data is already available to indicate the suitability of the material. Where such data is available it may not be necessary to determine the initial stability. However, in all cases ongoing stability checks are required if the material is to be used beyond the time span of the initial stability determination or stability data already available.

When testing for stability the following issues need to be considered with respect to the material to be tested, the parameters being measured and the minimum shelf life required:

(a) Selection of measurement method(s): since stability testing may need to be carried out over a long period of time (possibly years), it is necessary to select a method that provides precise measurements over a long period of time i.e. it is necessary to ensure that variations between one measurement and another made at a later time on the RM are not due to the calibration system used. It should be noted, however, that reproducibility data will almost invariably be less precise than repeatability data.

Two different approaches could be adopted:

- the system could be calibrated against a calibrant that can be reproduced at some time in the future. This may take the form of a standard calibration solution that is gravimetrically prepared or, if available, a CRM that is sufficiently stable could be used.
- where it is not possible to reproduce a calibrant over the time span required, the RM can be compared to a sample of that RM that has been stored at low temperatures such as -20°C or -70°C. In this case, it is assumed that at such low temperatures the RM will remain stable.

In some cases it may be possible to assess the stability of the material by measuring a different property value.

Example :

A single organic compound RM may be assessed for its purity. However, in order to continually assess the stability of the material, you can make use of a purity profile to indicate any deterioration in the material over time. Techniques such as HPLC and GLC should be used to identify and monitor any increase in breakdown products. It is generally easier to look for large changes in small things rather than small changes in large things. In addition, a moisture content determination should be carried out. In some cases, pH may adversely affect the stability of a material and should also be considered in such an assessment.

(b) Selection and number of samples: provided the material has been shown to be homogeneous, then any one unit or sub-sample (assuming it is not smaller than the sample size used for the homogeneity test) can be considered to be representative. Hence there is no constraint on the number of samples needed, or any need to randomly or systematically choose them. But, there will be a variation in results depending on the repeatability of the analytical technique. Thus replicate analysis should be carried out (the number of replicates depending on the precision of the method - if this is not known at least five should be used).

(c) Testing conditions: these will depend on the nature of the material and the level of knowledge already available on the stability of the material. The effects of the following must all be considered, in order to assess the appropriate storage conditions for the material and its lifespan:

- light
- moisture
- heat
- time

If no data exists then an initial stability trial is often conducted under accelerated conditions to enable an assessment of the material's suitability to be made within a relatively short time period (2 - 3 months)

i.e. the rate of any decomposition reaction which occurs at ambient temperature will generally be increased by a factor of two for every ten degrees rise in temperature - (Arrhenius equation).

Such initial accelerated stability testing is normally conducted on materials that have been stored at four approximate temperatures:

- -20°C (i.e. in a freezer)
- 4°C (i.e. in a refrigerator)
- 20°C (i.e. normal laboratory temperature)
- 40°C (i.e. in a laboratory oven).

However, such accelerated stability trials whilst giving an indication of future stability are not always reliable and thus it is necessary to undertake on-going stability testing. Conversely, such studies can sometimes give false negative results at elevated temperatures (i.e. degradation occurs), but the material can, in fact, be perfectly stable for a significant period at ambient temperatures.

Another fundamental point to note is that if a batch of material has been checked for stability for two years and is shown to be stable, it does not mean that it will necessarily be stable for another two years. On-going monitoring is required. Equally, if an identical batch of material is made, then it is reasonably to assume that it will be stable for at least two years prior to undertaking any stability monitoring.

Example :

Subjecting a food RM to accelerated conditions may indicate that it is stable for up to five years under normal laboratory conditions. However, in reality certain bacteria or moulds may form in much less time which will cause the material to deteriorate. This could be avoided by reducing the water content of the material to less than 5% (by mass), or gamma irradiation.

For long term stability trials i.e. after the material has been fully prepared and packaged, it would usually only be appropriate to monitor units of the RM stored under their normal storage conditions (e.g. the elevated temperature storage should be omitted). However, if it is envisaged that the reference material might also be

used in hot climates then clearly elevated temperature monitoring should also be continued. In terms of moisture and light, a range of appropriate conditions would also need to be assessed.

Where the only data available relates to a pure or purified substance or one with a different purity profile, care should be taken since, in reality, trace impurities present in the RM may change its long term stability.

Example:

A glass RM will usually be very stable (although migration of ions can occur within the matrix under certain conditions e.g. if used as an Scanning Electron Microscope standard) and in this case stability testing for the effect of heat (assuming that the RM is being stored at normal ambient temperatures) is not necessary. However, the glass could be hygroscopic and since this would affect the integrity of the material, stability testing should be conducted under a range of humidity conditions over various intervals of time.

Example:

For a food RM intended for the analyses of vitamins, in addition to testing the material stored under a range of different temperature conditions it would also be necessary to assess the stability under various degrees of light (since the vitamins may be prone to photolytic degradation).

Example:

A non-ferrous alloy RM, intended for the measurement of elemental content could be expected to be stable almost indefinitely and thus stability testing would not be necessary.

Example:

The stability of a pesticide may change dramatically depending on the pH of the formulation.

(d) Frequency of testing: this will be very dependent on the nature of the material. The initial accelerated stability testing would normally be conducted over a period of 3 to 6 months. Clearly, if the RM is to be prepared in batches which last for less than six months then stability would be established to the maximum length of time that the RM will be in use.

The recommended frequency of testing for the initial stability trial is 0, 1, 2, 3 & 6 months.

For long term stability trials the recommended frequency of testing is 0, 3, 6 and 12 months and then on an annual basis depending on the material.

In all cases it will depend on the intended length of time for which the RM is to be used. If reliable data is available i.e. for commercial materials or from previous stability trials on similar materials, the checks can be on a less frequent basis.

(e) Statistical tests to apply: it is beyond the scope of this document to give full mathematical formulae for the individual statistical tests to be applied. However, such formulae are readily available in numerous texts on

statistics including a number of the references given below. The basic concept is that the statistical tests need to assess whether there is any statistically significant variation between:

- the measured value of the RM at time (x) and that measured initially i.e. at time (zero).
This can be either absolute or relative to a suitable CRM.
- the measured value of the RM at time (x) and that on a sample of the RM stored at -20°C/-70°C

A recommended test would be the use of a t-test, normally conducted at the 95% confidence level, on the measurement results following the removal of statistical outliers (i.e. Cochran's, Grubb's and/or Dixon's tests could be used). However, this is not the only statistical test that could be applied, it is clearly situation dependent, the important point being that the test used fulfils the basic concept given above.



ISO Guide 31, (1981) : *Contents of certificates of reference materials*



ISO Guide 34, (1996) : *Quality system guidelines for the production of reference materials*



ISO Guide 35, (1989) : *Certification of reference materials - General and statistical principles*



ISO 5725-2, (1994) : *Accuracy (trueness & precision) of measurement methods - Basic method for the determination of repeatability and reproducibility for a standard measurement method*



European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials*



National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*



VAMSTAT (1994) - a computer based training package on statistics for analytical chemists

9. MEASUREMENT OF REFERENCE VALUE

When measuring the reference value the following issues need to be considered in the light of the material to be tested, the parameters being measured, the end use of the material and the measurement approach to be used:

(a) Selection of measurement method(s): a number of alternative approaches are available depending on the end use of the material:

- if the RM is to be used as a quality control check sample, the methods that will be controlled by the RM should be used. In this case it may not be necessary to obtain an accurate reference value since the RM will be used to compare the value obtained using the method for each subsequent batch of analyses.
- if the RM is to be used for assessing the performance of an analytical system then a more rigorous approach is required in order to obtain an accurate reference value. In this case it should be referenced against a CRM or the calibrated measuring device or instrument being used.
- if a CRM does not exist to which the RM can be referenced or the measurement method or measuring device is not certified then the reference value needs to be obtained by:

- ♦ a definitive (primary) method
- ♦ two or more methods and preferably include some independent check
- ♦ an interlaboratory exercise involving a reasonable number of participants

If a number of laboratories already take part in some form of regular interlaboratory exercise (e.g. between various sites of one company) this may offer a mechanism for measuring the RM using a number of methods and laboratories. This would be particularly relevant where no appropriate CRM exists, the measurement method or measuring device is not certified or there are no better methods available.

Example:

For a food RM, intended as a check sample for the determination of trace metals, providing it has been proved to be homogeneous and stable, it would be sufficient to reference it against a suitable CRM using the method that the RM will be used to control. However, if the RM was to be used for performance assessment, perhaps where a suitable CRM was not available, it would need to be measured either by using two or more methods or a definitive method such as neutron activation analysis or by an interlaboratory exercise.

In all cases care must be taken when selecting the method to use to obtain the reference value to ensure that the measurement made is valid for the intended purpose of the RM and indeed in many cases it may be necessary to employ more than one method.

Example :

For a single organic compound RM the reference value will often be in the form of a purity level which may have been obtained using some chromatographic technique. However, the following are typical errors that can occur:

- the purity is calculated by subtracting the level of impurities found from 100%, but this will only be accurate if all impurities have been found.
- using only one method such as high performance liquid chromatography may not assess all impurities since it will not detect such things as moisture, the main peak may mask certain impurities and it does make the assumption that the response rate is the same for all impurities detected.
- differential scanning calorimetry is used for purity determination, but may sometimes not give correct results e.g. it will not detect impurities that form solid solutions or the presence of chiral compounds.

Therefore a number of different methods are required to ensure that the purity has been appropriately assessed and the limitations of each method taken into account.

Example :

A titrimetric method may be used to obtain the reference value of the RM produced which under many conditions provides an absolute measurement technique. However, depending on the nature of the material and the type of titration being used, interfering substances may be present in the matrix which can render the result obtained unreliable.

(b) Selection and number of samples: provided the material has been shown to be homogeneous, then any one unit or sub-sample (assuming it is not smaller than the sample size used for the homogeneity test) can be considered to be representative. Hence there is no constraint on the number of samples needed, or any need to randomly or systematically choose them. However, there will be a variation in results depending on the repeatability of the measurement technique. Replicate analysis should therefore be carried out. The number of replicates to use depends on the precision of the method being used. If this is not known, it is recommended that at least ten replicate analyses be carried out where only one method is used and five replicates for each method, when more than one method is used. Measurements made under reproducibility conditions will usually show a greater variation than those made under repeatability conditions.

(c) Frequency of measurement: the reference value of the RM should be checked whenever there is an unexpected result obtained during its regular use, if it has been exposed unavoidably to abnormal conditions e.g. the refrigerator breaks down overnight, or on a periodic basis (say one to six months) depending on the nature of the material. Clearly a glass RM would need to be checked on a less frequent basis than a microbiological RM (see section 8 on RM stability testing).

(d) Statistical tests to apply: it is beyond the scope of this document to give full mathematical formulae for the individual statistical tests to be applied. However, such formulae are readily available in numerous texts on statistics including a number of the references given below. The exact statistical tests to apply will depend on the type of referencing method used but some guidance is given below:

- referencing against a CRM: the CRM is used to calibrate the measurement system that is used to undertake replicate analysis on the RM. The mean and confidence interval of the replicate measurements can be calculated following the elimination of statistical outliers (i.e. Cochran's, Grubb's and/or Dixon's tests could be used). In some cases a good quality RM may be available and this may equally be used to calibrate the measurement system.

- one or more methods or an interlaboratory exercise : the results obtained are assessed for variance using a suitable ANOVA (analysis of variance) test following the elimination of statistical outliers (tests given above). The mean and confidence interval can then be calculated.

- definitive method : the mean and confidence interval can be calculated following the elimination of statistical outliers (tests given above).



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ISO 5725-2, (1994) : *Accuracy (trueness & precision) of measurement methods - Basic method for the determination of repeatability and reproducibility for a standard measurement method*



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VAMSTAT (1994) - a computer based training package on statistics for analytical chemists

10. DOCUMENTATION

It is essential that records are kept for each stage of the in-house production of a RM. Such records should at least contain the following information:

- ❖ Name(s) / Reference number(s) /Batch number(s) of the materials to which the documentation applies
- ❖ Origin/source of the material(s)
- ❖ Details on the material preparation techniques used
- ❖ Dates of all measurements made
- ❖ Details of analytical methods used including calibration i.e. CRMs used
- ❖ Raw/primary data
- ❖ Reasons for rejection of any data on technical grounds
- ❖ Results of statistical analysis and the methods used
- ❖ The reference value(s) to use
- ❖ Name(s) of analysts who were responsible for the material preparation, analytical measurements and the calculation & interpretation of the data

In addition, details on stability, homogeneity, storage requirements and instructions on the correct use of the reference material should be noted.

All documents should be retained as specified in your own Organisation's quality system.



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ISO Guide 34, (1996) : *Quality system guidelines for the production of reference materials*



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National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*

11. POST PRODUCTION

The RM should be stored under the appropriate conditions that have been derived either from reliable and relevant available data or from stability tests carried out (see Section 8). It may be prudent to store part of the batch under conditions that are appropriate for daily use and the rest of the batch under more stringent conditions to enhance the overall lifespan of the whole batch.

Example:

For a food RM which is split into a number of 50g units, each unit lasting for about a month, it would be satisfactory and indeed beneficial to store the current unit being used at 4°C (i.e. it has been assessed to be stable for at least a month at this temperature) and the rest of the batch at -20°C (i.e. the material is much more stable at the lower temperature). This storage regime will increase the lifespan of each batch of material, thus allowing more cost effective larger batches to be produced each time.

If the RM is to be stored and used over a long period of time (the length of time obviously depends on the nature of the material) then periodic checks for long term stability need to be made (see Section 8). If reliable and relevant data are available i.e. for readily available materials or from previous stability trials on similar materials, the checks can be made less frequently.

When a new batch of the RM is prepared, the new batch should be compared against the old batch to ensure continuity (e.g. impurity profiles, moisture content, etc.).



ISO Guide 33, (1989) : *Uses of certified reference materials*



ISO Guide 34, (1996) : *Quality system guidelines for the production of reference materials*



European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials*



National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*

APPENDIX A

HOW TO OBTAIN LISTED REFERENCES

1. Standards and Guides published by the International Standardisation Organisation (ISO) and the British Standards Institute (BSI) may be obtained from :

British Standards Institute
389 Chiswick High Road
London
W4 4AL

Tel : 0181 996 7000
Fax : 0181 996 7001

2. European Commission Document, BCR/48/93 (Dec 1994) : *Guidelines for the production and certification of BCR reference materials*

- available from : European Commission
DG XII
Standards, Measurements and Testing Programme
Rue de la Loi, 200
B-1049 Brussels
Belgium

Fax : 00 32 2 295 80 72

3. National Institute of Standards and Technology Publication 260-100 (1993) : *Standard Reference Materials - handbook for SRM Users*

available from : National Institute of Standards and Technology
Standard Reference Materials Programme
Building 202, Room 204
Gaithersburg
MD 20878-9950
USA

Fax : 00 1 301 948 3730

4. International Harmonised Protocol for Proficiency Testing of (Chemical) Analytical Laboratories - Thompson & Wood

published in : Journal of AOAC International : Vol.76, No.4, 1993

5. General Principles of Good Sampling Practice - Crosby.N.T & Patel.I

available from : The Royal Society of Chemistry
Turpin Distribution Services Limited
Blackhorse Road
Letchworth
Herts, SG6 1HN

Tel : 01223 420066

Fax : 01223 423623

6. VAMSTAT (1994) - a computer based training package on statistics for analytical chemists

available from the Royal Society of Chemistry - details above